

Modeling two-dimensional magnetic resonance measurements in coupled pore systems

L. M. Schwartz,¹ D. L. Johnson,¹ J. Mitchell,^{2,3} T. C. Chandrasekera,² and E. J. Fordham³

¹*Schlumberger Doll Research, One Hampshire Street, Cambridge, Massachusetts 02139, USA*

²*Department of Chemical Engineering and Biotechnology, University of Cambridge, Cambridge CB2 3RA, United Kingdom*

³*Schlumberger Gould Research, High Cross, Cambridge CB3 0EL, United Kingdom*

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We present numerical simulations of a two-dimensional (2D) nuclear magnetic resonance process, T_2 -storage- T_2 , on a simple mixed porosity system, the micrograin consolidation (μ GC) model. The results of these calculations are compared with predictions based on the analytic two-site exchange model, for which we have independently established numerical values for all the input parameters. Although there is qualitative and semiquantitative agreement between the two models, we identify specific instances where the two-site model fails to properly describe the combined effects of relaxation and diffusion. Generally, these instances occur when a gradient in magnetization within the large pores of the μ GC model is established during the initial phase of the 2D process. The two-site model assumes that the magnetization is spatially uniform within each of its subpore systems and thus cannot describe such effects.

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I. INTRODUCTION

Two-dimensional (2D) nuclear magnetic resonance (NMR) relaxation experiments have been applied to a wide range of fluid-filled porous media including petroleum reservoir rocks [1–7], bead packs [8], cements [9,10], and gypsum plaster [11]. In addition, they have been used to study catalysts [12], surfactants [13], liquid crystals [14,15], and other materials of interest [16,17].

Two-dimensional NMR relaxation measurements provide a correlation between the initial (encoding) and final (detection) parts of the experiment. The encoding and detection sequences may involve longitudinal T_1 or transverse T_2 decay. An important class, including T_2 - T_2 [6,10,18] and T_1 - T_1 [19] correlations, incorporates a user-controlled storage interval t_s between the encoding and detection intervals. Typically, the experimental data are processed to produce a 2D distribution of lifetimes. In the resulting cross plot off-diagonal peaks may be observed; these have been interpreted, heuristically, in terms of diffusive exchange between distinct spin populations. Indeed, by varying t_s , exchange rates or coupling parameters have been estimated by several authors [6–8,10,19]. While the well-defined storage time is appealing for qualitative interpretation, there is a continuing controversy as to the link between the off-diagonal peak amplitudes and t_s .

In this paper we focus on the T_2 - t_s - T_2 correlation in model systems with distinct pore subpopulations. Our calculations proceed in the time domain; we follow the spin decay from the initial encoding sequence through the storage period and into the detection sequence. Our results are easily displayed as 2D distributions of relaxation lifetimes and we can directly monitor the dependence of the off-diagonal peaks amplitudes on the storage time t_s . The first model we work with is a two-site exchange model introduced by Zimmerman and Brittin [20] and by McConnell [21]. The model is based on several critical assumptions, most important that each spin population, in isolation, exhibits just a single relaxation mode. Physically, this would be consistent with very weak surface relaxation as discussed by Brownstein and Tarr [22]. However, individual pores in real samples may exhibit multiple

relaxation modes [22]. Such behavior can arise for a variety of reasons, for example, from intermediate or large values of surface relaxation or from magnetic susceptibility contrast inducing intrapore field gradients [6,23]. We turn next to the micrograin consolidation (μ GC) model. This model is based on an explicit three-dimensional pore geometry that idealizes the mixture of macropores and micropores found in many carbonate grainstones [24]. Such textures are important in oil reservoirs and have been studied experimentally using the T_2 - T_2 correlation [7].

The purpose of our analysis is to clarify the interpretation of T_2 - t_s - T_2 measurements. Since the value of 2D measurements is related to the way that they describe coupling between distinct pore populations, it will be useful to modulate this coupling directly in our calculations. This is easily done in both the two-site exchange and μ GC models and we make extensive use of this feature. In the first case we can turn off (i.e., set to zero) the coupling parameters during any of the three stages of the measurement. Similarly, in the μ GC model we can introduce a barrier between the macropores and micropores. As in the two-site exchange model, the barrier can be put in place during the spin encoding, spin storage, or spin detection parts of the calculation. With the ability to monitor the coupling, we show that while the amplitude of the off-diagonal peaks in the T_2 - T_2 map is initially proportional to t_s , it is not true that these peaks are wholly controlled by diffusive interaction during the storage period. Diffusion within a given pore may already impose a magnetization gradient even during the first (encoding) phase of the T_2 - t_s - T_2 protocol.

Section II of this paper has three parts, a brief review of NMR diffusion and surface relaxation, a description of the experimental protocol, and a summary of the equations underlying the two-site exchange and μ GC models. In Sec. III the results of our calculations are presented. In many respects the two-site exchange and μ GC models yield correlations that are quite similar. However, when we modulate the diffusive coupling some interesting differences between the two models are revealed. In Sec. IV we analyze our results based on the normal modes of the diffusion equation. We show that there

is a significant simplification in the formalism if one assumes that the system can be approximated as a linear combination of only two different basis functions. One of the problems in applying the two-site exchange model to real experimental data is that it is difficult to unambiguously determine values for the parameters that define the model. Here, because we have control over the μ GC geometry, we are able to deduce all the parameters cleanly. We then compare the predictions of the two-site model directly against our simulations of this 2D process. We conclude (Sec. V) by summarizing the relationship between the two-site exchange and μ GC models. Under certain conditions the two-site exchange model can be relied upon to estimate exchange rates. However, more generally, direct solutions of the diffusion equation in specific geometries are required, as is done in a variety of different contexts with existing 3D microtomography techniques [25].

II. BASIC CONSIDERATIONS

A. Diffusion and magnetic relaxation

Before considering the 2D NMR measurement, it will be useful to summarize the equations that describe magnetic resonance in fluid-saturated porous media. We picture a system initially prepared with a uniform magnetization $M(\mathbf{r}, t = 0) = M_0$ in the fluid phase. Here M_0 is the initial magnetic moment per unit pore volume in the sample. The evolution of $M(\mathbf{r}, t)$ is then controlled by a combination of molecular diffusion within the pore space and absorption at the pore-grain interface. The basic equations are discussed in detail by Brownstein and Tarr [22]. The evolution of the magnetization [magnetic moment per unit volume $M(\mathbf{r}, t)$] is described by the diffusion equation

$$D_0 \nabla^2 M(\mathbf{r}, t) = \frac{\partial M(\mathbf{r}, t)}{\partial t} \quad (1)$$

subject to the boundary condition on the interface between the pore fluid and the solid grain

$$[D_0 \hat{\mathbf{n}} \cdot \nabla M(\mathbf{r}, t) + \rho M(\mathbf{r}, t)]_{\text{interface}} = 0. \quad (2)$$

[We have set the bulk fluid relaxation equal to zero, without loss of generality, because this effect can be added simply by the substitution $M(\mathbf{r}, t) \rightarrow e^{-t/T_B} M(\mathbf{r}, t)$, where T_B is either the longitudinal or transverse relaxation time, as the case may be [22].]

Here D_0 is the diffusion coefficient for the bulk fluid and the parameter ρ describes the interface absorption strength. In the present analysis the values of ρ will be assumed constant over the entire interface [26]. Given these equations, the total magnetic moment of the sample can be represented as a standard eigenfunction-eigenvalue expansion

$$M(t) = M_0 V_P \sum_{n=1}^{\infty} A_n e^{-\lambda_n t}, \quad (3)$$

where V_P is the volume of the pore space. Each decay rate λ_n is an eigenvalue pertaining to Eqs. (1) and (2); each amplitude A_n is given in terms of the associated eigenmode $\mathbf{q}_n \equiv q_n(\mathbf{r})$. These are defined via solutions to the eigenvalue-eigenvector problem

$$D_0 \nabla^2 q_n(\mathbf{r}) + \lambda_n q_n(\mathbf{r}) = 0, \quad n = 1, 2, 3, \dots, \quad (4)$$

subject to the same boundary condition (2). The set $\{\mathbf{q}_n\}$ forms a complete and orthonormal basis for continuous functions of position within the pore space, properties that we shall use later. The amplitudes in Eq. (3) are

$$A_n = \frac{(\mathbf{m}_0 \cdot \mathbf{q}_n)^2}{V_P}, \quad (5)$$

where $m_0(\mathbf{r})$ is a function that is equal to unity in the pore space and zero elsewhere. Our notation is that the inner product of two functions is the usual

$$\mathbf{f} \cdot \mathbf{g} \stackrel{\text{def}}{=} \int_{V_P} f(\mathbf{r}) g(\mathbf{r}) d^3 \mathbf{r}, \quad (6)$$

where $f(\mathbf{r})$ and $g(\mathbf{r})$ are any two functions of position within the pore space. Equations (1)–(6) are discussed in detail in Ref. [22].

In general, the eigenfunctions $q_n(\mathbf{r})$ and eigenvalues λ_n depend on the value of ρ and on the shape of the pore-grain interface. In the two most familiar experiments one looks at the decay of either longitudinal (1) or transverse (2) magnetization. The above equations describe both measurements although, typically, $1.5 < \rho_2/\rho_1 < 2$.

A word on notation is in order: For the purposes of equations that involve the inner product it is convenient to represent a state function in bold type, e.g., $\mathbf{f}$. However, this represents a continuous function of position within the pore space, e.g., $f(\mathbf{r})$. So $\mathbf{f}$ and $f(\mathbf{r})$ are the same thing. Later, in Sec. IIC we introduce the two-site exchange model in which any state of the system is represented by a two-dimensional vector $\mathbf{f} = (f_A, f_B)$. Here $\mathbf{f}$ represents a function that takes on the constant value f_A in pore A and the constant value f_B in pore B . The inner product between two so-defined state vectors $\mathbf{f}$ and $\mathbf{g}$ still follows from the same definition (6): $\mathbf{f} \cdot \mathbf{g} = f_A g_A V_A + f_B g_B V_B$, where V_A and V_B are the volumes of pore A and pore B , respectively. The equations that we presented above, as well as those we develop in Secs. IIB and IV B, are equally applicable to the formalism in which the magnetization is described by continuous functions or to that in which it is described by two-dimensional vectors. In the latter case the integration over the pore space becomes trivially simple.

B. The 2D measurement protocol

It will be useful to summarize the experimental protocol to which our calculations relate. The first T_2 interval comprises a standard Carr-Purcell-Meiboom-Gill (CPMG) series of N_1 spin echoes [27] (or see [28]). The duration of this preparation portion is $\tau_1 \equiv N_1 t_e$, where t_e is the temporal separation of individual spin echoes. At the end of τ_1 , a 90° radio frequency (rf) pulse is applied to store the encoded spins on the longitudinal axis for a time t_s , during which the spins undergo longitudinal relaxation. The spins are then restored to the transverse plane by another 90° rf pulse and a second CPMG sequence of N_2 echoes is used to monitor the decay of magnetization. The duration of this second T_2 interval is $\tau_2 \equiv N_2 t_e$. Data (spin echo amplitudes) are acquired only in this latter portion of the experiment, but a 2D data matrix is constructed by varying the number of echoes N_1 in the first T_2 interval. In this manner, the experiment is sensitive to the decay of magnetization in the first T_2 interval via the amplitude of the

first echo in the second T_2 interval. The entire rf pulse sequence is illustrated in Fig. 1(a). We denote the magnetization at an arbitrary time in the experiment by $M(\tau_2, \tau_1; t_s)$.

A 2D relaxation time correlation is obtained by numerical inversion of the 2D Fredholm integral that describes the expected form of the data; methods for inverting NMR relaxation and diffusion correlations are reviewed elsewhere [29]. The relevant integral equation is

$$M(\tau_2, \tau_1; t_s) = \iint K(\tau_2, T_2^{(1)}, \tau_1, T_2^{(2)}) \times P(T_2^{(1)}, T_2^{(2)}; t_s) d \log_{10} T_2^{(1)} d \log_{10} T_2^{(2)} + e(\tau_2, \tau_1), \quad (7)$$

where K is the kernel function (expected form of the data), P is the required distribution, and e is experimental noise. The different T_2 axes are denoted by $T_2^{(2,1)}$; $T_2^{(1)}$ is associated with the τ_2 data dimension for consistency with the familiar one-dimensional lifetime distribution. For the T_2 - T_2 experiment

$$K(\tau_2, T_2^{(1)}, \tau_1, T_2^{(2)}) = \exp(-\tau_2/T_2^{(1)}) \exp(-\tau_1/T_2^{(2)}). \quad (8)$$

To solve Eq. (7) for the distribution $P(T_2^{(1)}, T_2^{(2)}; t_s)$ we employ Tikhonov regularization [30]; a stable solution is obtained in the presence of noise by forcing the resultant distribution to be non-negative, smooth, and bounded by finite values of $T_2^{(1,2)}$ [31–34].

The theoretical description of this experiment follows along the lines of the derivation of Eqs. (3) and (5). Between $t = 0$ and $t = \tau_1$ the magnetization is given by expressions analogous to Eqs. (3) and (5) using the transverse eigenfunctions and eigenvalues $\{\lambda_n, \mathbf{q}_n\}$. Then $M(\mathbf{r}, \tau_1)$ is the initial condition for an expansion based on the longitudinal eigenfunctions and eigenvalues $\{\mathbf{Q}_l\}$ and μ_l . [Note that, because $M(\mathbf{r}, \tau_1)$ will in general not be spatially uniform, its projection onto the longitudinal eigenfunctions is nontrivial.] Finally, $M(\mathbf{r}, \tau_1 + t_s)$ is the initial state for the final transverse basis function expansion between $t = \tau_1 + t_s$ and $\tau_1 + t_s + \tau_2$. The result is [31,33]

$$M(\tau_1, \tau_2; t_s) = \sum_{m,n} \mathcal{M}_{nm}(t_s) e^{-[\lambda_n \tau_1 + \lambda_m \tau_2]}, \quad (9)$$

where

$$\mathcal{M}_{nm}(t_s) = M_0 \sum_l [\mathbf{m}_0 \cdot \mathbf{q}_n][\mathbf{q}_n \cdot \mathbf{Q}_l][\mathbf{Q}_l \cdot \mathbf{q}_m] \times [\mathbf{q}_m \cdot \mathbf{m}_0] e^{-\mu_l t_s}. \quad (10)$$

From Eq. (10) we see, by inspection, that the matrix is symmetric: $\mathcal{M}_{nm}(t_s) = \mathcal{M}_{mn}(t_s)$. (Here and for the remainder of this article, l, m, n are indices relating to the specific normal modes of the problem and are not to be confused with echo numbers, mentioned earlier.) In practical terms we take it that any well-defined narrow peak in the experimentally determined distribution $P(T_2^{(1)}, T_2^{(2)}; t_s)$ at values $T^{(n)}$ and $T^{(m)}$, say, represents a broadened δ function in the space of the eigenvalue decay rates $\lambda_n = 1/T^{(n)}$ and $\lambda_m = 1/T^{(m)}$.

As mentioned earlier, the T_1 normal modes form a complete set, which we write in dyadic notation as

$$\sum_l \mathbf{Q}_l \mathbf{Q}_l = I \quad (11)$$

and similarly for the T_2 modes

$$\sum_l \mathbf{q}_l \mathbf{q}_l = I. \quad (12)$$

[When Eqs. (11) and (12) are applied to continuous functions of position, $I = \delta(\mathbf{r} - \mathbf{r}')$; for discrete basis sets, $I = \delta_{ij}$.] Moreover, the basis functions are orthonormal:

$$\mathbf{Q}_n \cdot \mathbf{Q}_m = \delta_{nm} \quad \forall n, m \quad (13)$$

and

$$\mathbf{q}_n \cdot \mathbf{q}_m = \delta_{nm} \quad \forall n, m, \quad (14)$$

where δ_{nm} is the Kronecker delta. These general properties allow one to derive the limiting behavior for short storage times $\mu_l t_s \ll 1$. All the exponentials in Eq. (10) are equal to unity, the sum on l collapses because of Eq. (11), and the sum on m collapses because of Eq. (14) with the result that

$$M(\tau_1, \tau_2; t_s \rightarrow 0) = M_0 \sum_n |\mathbf{q}_n \cdot \mathbf{m}_0|^2 e^{-\lambda_n(\tau_1 + \tau_2)}. \quad (15)$$

However, this is just Eq. (3) with $t = \tau_1 + \tau_2$. If the storage time is short enough the magnetization distribution does not change during storage and one has simple T_2 decay; the off-diagonal elements of $\mathcal{M}_{nm}$ vanish identically.

C. Models

We will compare calculations based on two models that can be used to describe the experiment represented in Fig. 1. Throughout this paper two assumptions are made to simplify our analysis. **First, we ignore both longitudinal and transverse relaxation in the bulk fluid.** As noted above, this effect can easily be added after calculating the surface controlled relaxation. Second, we set $\rho_1 = 0$ so there is no relaxation during the storage period. This maximizes the number of

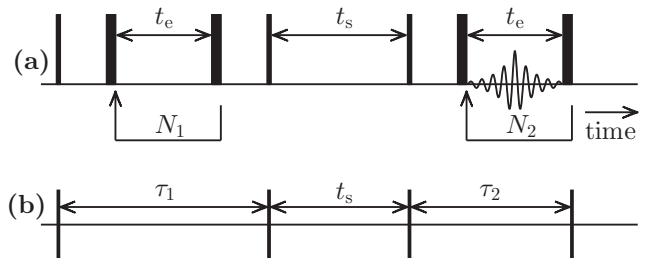


FIG. 1. Schematic of the T_2 - T_2 exchange measurement: (a) the rf pulse sequence diagram for the NMR experiment; thin and thick vertical lines represent 90° and 180° rf pulses, respectively. N_1 echoes are generated in the first T_2 encoding interval and N_2 echoes are monitored in the second T_2 interval. The spins are stored for a time t_s between the two T_2 intervals. The timing diagram (b) indicates the corresponding parts of our calculations. The duration of the first T_2 interval is τ_1 and the duration of the second T_2 interval is τ_2 . For a given t_s , τ_1 is varied on a logarithmic scale and the data are then processed.

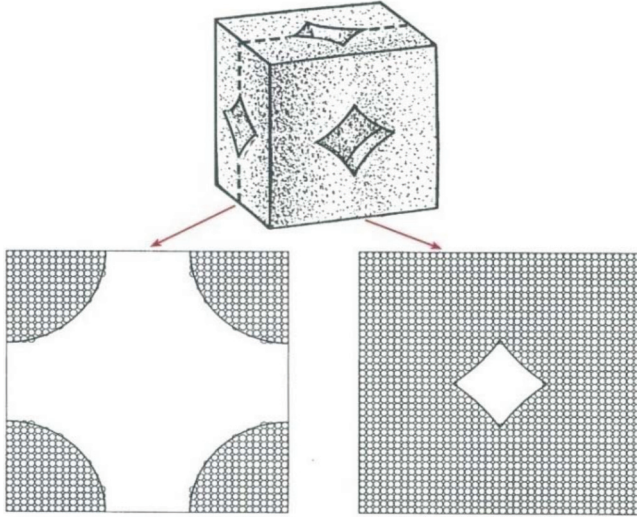


FIG. 2. Microporous version of the grain consolidation model in which the large grains are comprised of smaller grains whose centers lie on a lattice constructed by subdividing the original unit cell. The solid line separating the macropores and micropores represents a synthetic barrier that can be switched on and off during the random walk simulation.

spins that survive to the data acquisition segment and are available for coupling between distinct spin populations. This assumption simplifies our numerical simulations, but does not change the qualitative features of our results.

The μ GC model, shown in Fig. 2, is based on a three-dimensional periodic packing with a bimodal pore-size distribution [24,35]. Here large grains, each of which is itself porous, are packed on a simple cubic lattice. The large grains are comprised of smaller (possibly) overlapping grains centered on a lattice constructed by dividing the original unit cell into smaller cells. The scale factor governing this division controls the degree of mismatch in the model's pore sizes. If the large grains overlap, the μ GC model has pathways through the large open pores and (roughly) parallel pathways through the microporous grains. For the calculations described in this paper the edge of the large cube is taken to be $200 \mu\text{m}$ and the scale factor is 20, corresponding to small cells with an edge length of $10 \mu\text{m}$. The radius of the large grains R_M and that of the small grains R_m are given in Table I. Also given in the table are the corresponding macroporosities and microporosities (ϕ_M and ϕ_m , respectively). [The intrinsic porosity within the microporous region is $\phi_\mu = 0.180$ and $(1 - \phi_M)\phi_\mu = \phi_m$.] In Refs. [24,35] we have shown how variable step size random walks can be applied to the μ GC model to simulate the solution of Eqs. (1) and (2). These same techniques can be applied to the 2D measurement sequence described above. As in Ref. [35], the diffusion coefficient D_0 assigned to the fluid that fills the pore space is $1 \mu\text{m}^2/\text{ms} = 10^{-5} \text{ cm}^2/\text{s}$.

TABLE I. Parameter values for the simulations based on the μ GC model.

R_M	$125 \mu\text{m}$
ϕ_M	0.149
R_m	$6.1 \mu\text{m}$
ϕ_m	0.153
ϕ_{total}	0.302
D_0	$10^3 \mu\text{m}^2/\text{s}$
ρ_1	0
ρ_2	$2 \mu\text{m}/\text{s}$

The second model, which can be treated analytically, describes two spin reservoirs coupled by diffusive exchange. We picture two spin populations (in our case the macropores and the micropores) contributing signal volumes S_A and S_B , which are coupled to each other. Here $S_{A,B}$ represents the difference from the equilibrium values of the magnetic moment in A or B ; the total out-of-equilibrium magnetic moment is simply $M(t) = S_A(t) + S_B(t)$. In the absence of coupling, each population is characterized by a single rate for longitudinal (transverse) decay rate $R_A^{(2)} = 1/T_A^{(2)}$ and $R_B^{(2)} = 1/T_B^{(2)}$. [In the language of Eq. (3), a single mode describes the magnetization decay in the absence of coupling.] In the language of Ref. [22], each subsystem is assumed to be in the so-called fast diffusion limit in the sense that $\rho a/D_0 \ll 1$, where a is a characteristic spatial dimension of each subsystem. This assumption guarantees that the magnetization within each subsystem is spatially uniform at any time.

Analytic expressions for the time dependence of S_A and S_B during the $\tau_1 \rightarrow t_s \rightarrow \tau_2$ measurement sequence are derived by solving the coupled equations

$$\begin{pmatrix} \frac{dS_A}{dt} \\ \frac{dS_B}{dt} \end{pmatrix} = \begin{pmatrix} -R_A - \kappa_A & \kappa_B \\ \kappa_A & -R_B - \kappa_B \end{pmatrix} \cdot \begin{pmatrix} S_A \\ S_B \end{pmatrix}, \quad (16)$$

where the parameters κ_A and κ_B describe the effective exchange rates between the two reservoirs. When the magnetization densities in the two pores are equal $S_A/V_A = S_B/V_B$ (V_A and V_B are the volumes of the pores) exchange from $A \rightarrow B$ occurs at the same rate as $B \rightarrow A$. This condition of detailed balance implies $\kappa_A V_A \equiv \kappa_B V_B \stackrel{\text{def}}{=} k$.

Equation (16) describes a heuristic two-site model with potential applicability to a variety of situations. Here we explore its applicability to the evolution of spin populations in the microporous μ GC model. Thus we envision that the parameter k is related to the diffusivity D_0 , though the latter parameter does not appear explicitly in the model.

The magnetization in the coupled system is evaluated from the eigenvalues of Eq. (16) [9,33]:

$$\lambda_{\pm} = \frac{R_A + R_B + k(u_A + u_B) \pm \sqrt{(R_A - R_B)^2 + 2k(R_A - R_B)(u_A - u_B) + k^2(u_A + u_B)^2}}{2}, \quad (17)$$

where $u_{A,B} = 1/V_{A,B}$. For each eigenvalue $\lambda_{\pm}$ there is an associated eigenvector $(S_A^{\pm}, S_B^{\pm})$. We have found it convenient to express our results in terms of the magnetization $\mathbf{m}^{\pm} = (m_A^{\pm}, m_B^{\pm}) = (S_A^{\pm}/V_A, S_B^{\pm}/V_B)$. From Eq. (16) we may use either of the following expressions for the (unnormalized) normal modes:

$$\begin{aligned}\mathbf{m}^{\pm} &= [k, (R_A V_A + k - \lambda_{\pm} V_A)], \\ \mathbf{m}^{\pm} &= [(R_B V_B + k - \lambda_{\pm} V_B), k].\end{aligned}\quad (18)$$

The inner products that appear in Eqs. (5) and (10) are given by Eq. (6), which becomes

$$\mathbf{f} \cdot \mathbf{g} = f_A g_A V_A + f_B g_B V_B, \quad (19)$$

where $f_{A,B} = S_{A,B}/V_{A,B}$ is the excess magnetization density in pore *A* or *B* for mode *f* and similarly for mode *g*. The normalized normal modes for the two-site exchange model are therefore

$$\mathbf{q}_{1,2} = \frac{\mathbf{m}^{\mp}}{\sqrt{\mathbf{m}^{\mp} \cdot \mathbf{m}^{\mp}}}. \quad (20)$$

Similarly, the normal modes for the storage part $\mathbf{Q}_{1,2}$ are given by the same procedure, but with $R_A = R_B = 0$.

If there is no coupling between the two pore systems ($k = 0$) these decay rates become simply R_A and R_B . It will be useful to look at the limiting behavior of the eigenvalues as a function of the coupling parameter, k . For small k

$$\lim_{k \rightarrow 0} \lambda_{\pm}(k) = \begin{cases} R_A + k/V_A + O(k^2) \\ R_B + k/V_B + O(k^2), \end{cases} \quad (21)$$

while for large k

$$\begin{aligned}\lim_{k \rightarrow \infty} \lambda_{+}(k) &= k(u_A + u_B) + \frac{R_A u_A + R_B u_B}{u_A + u_B} + O(1/k), \\ \lim_{k \rightarrow \infty} \lambda_{-}(k) &= \frac{R_A V_A + R_B V_B}{V_A + V_B} + O(1/k).\end{aligned}\quad (22)$$

A useful estimate of the crossover value k_{xc} separating the weak coupling regime and the strong coupling regime is given by the intersections of either of the two pairs of asymptotes in Eqs. (21) and (22). As it happens, these intersections occur at the same value of k , which is

$$k_{xc} = \frac{|R_A - R_B|}{u_A + u_B}. \quad (23)$$

The eigenvalues defined by Eq. (17) and their limiting values (21) and (22) are plotted as functions of k in Fig. 10 for specific choices of the parameters $R_{A(B)}$ and $V_{A(B)}$.

Our purpose is to clarify the interpretation of T_2 - t_s - T_2 measurements by comparing calculations based on the two-site exchange and μ GC models. Since the value of 2D measurements is related to the way that they describe coupling between distinct pore populations, it will be useful to modulate this coupling directly in our calculations. This is easily done in both models. In the two-site exchange model we can turn off (i.e., set to zero) the coupling parameters $\kappa_{A(B)}$ during any of the three stages of the measurement. Similarly, in the μ GC model we can introduce a *barrier* (Fig. 2) between the macropores and micropores. As in the two-site exchange model, the barrier can be put in place during the spin preparation τ_1 , spin storage t_s , or data acquisition τ_2 parts of the measurement sequence.

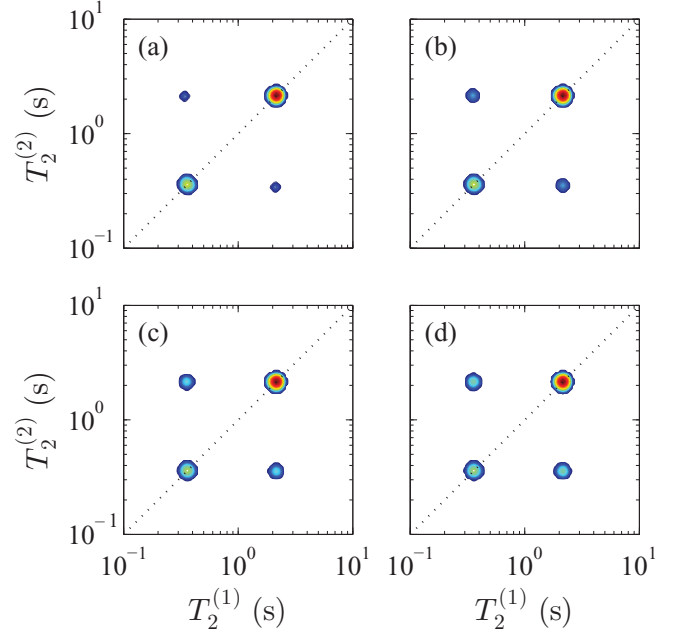


FIG. 3. (Color online) The T_2 - T_2 correlation plots generated using the two-site exchange model. The set of parameter values is given in Table II. Exchange is allowed during all three intervals in the calculation. The storage times are (a) $t_s = 100$ ms, (b) 400 ms, (c) 700 ms, and (d) 1000 ms. The dotted line indicates the diagonal $T_{2(1)} = T_{2(2)}$ in each plot. The apparent relaxation rates are $\lambda_1 = 0.44 \text{ s}^{-1}$ and $\lambda_2 = 2.82 \text{ s}^{-1}$.

III. CALCULATIONS

In Figs. 3 and 4 we show representative calculations based on the two-site exchange and μ GC models. The T_2 - T_2 cross plots are shown for four values of the storage time t_s . Here the parameter values used in the two-site exchange model have been chosen so that the results approximate those of the μ GC model. We describe how this is done in Sec. IV A. The parameter values are listed in Table II. In these calculations the model parameters have been adjusted to give roughly equal volumes in the two pore populations and, as noted above, we have set $\rho_1 = 0$. While the storage of the spins during t_s is perfect, there is nevertheless significant rearrangement of the spin density because the boundary condition at the pore-grain interface has changed. [As expected, calculations with $\rho_1 \neq 0$ (e.g., $\rho_1 = \rho_2/2$) yield results that are similar to those shown on Figs. 3 and 4, but the off-diagonal peaks are less pronounced.]

Generally, the diagonal peaks in the T_2 - T_2 cross plots are thought to represent the lifetime spectrum within distinct pore populations and the off-diagonal peaks represent diffusive coupling between them. Indeed, the off-diagonal peaks are often referred to as *exchange* peaks and are believed to represent diffusion (between the two populations) that takes place during the storage period t_s . It is well-established [31,33] that this is not really true; according to Eq. (10), the off-diagonal peaks are controlled by the degree to which the normal modes in the storage period are different from those in the initial T_2 decay. In principle, one may have off-diagonal peaks corresponding to different modes within the same single pore. In the very special case that the T_2 modes are describable

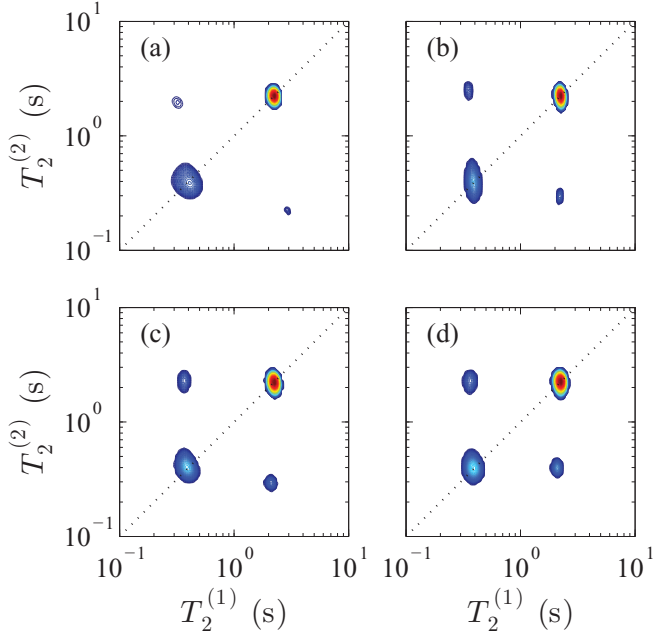


FIG. 4. (Color online) Four T_2 - T_2 cross plots based on the μ GC model. We have taken $\rho_1 = 0$ and four values of t_s : (a) $t_s = 100$ ms, (b) 400 ms, (c) 700 ms, and (d) 1000 ms. The dotted line indicates the diagonal $T_{2(1)} = T_{2(2)}$ in each plot. In these calculations the model parameters have been arranged so that the microporosity $\phi_m = 0.16$, the macroporosity $\phi_M = 0.14$, and the transverse surface relaxation parameter $\rho_2 = 2.0 \mu\text{m/s}$.

as being weakly coupled in the sense of Eqs. (21), the term “exchange” may have some approximate validity, but even so it is really just a manner of speaking. In physical terms, of course, diffusive exchange between the two pore populations is taking place over the entire measurement sequence. To better understand what is happening, it will be useful to turn the coupling on and off during the different stages of the model calculations. As we noted above, this can be done by setting the coupling parameter $\kappa \rightarrow 0$ in the two-site exchange model and by imposing a barrier between the macropores and micropores in the μ GC model.

In Figs. 5 and 6 four distinct protocols are employed to compare calculations based on the two-site exchange and μ GC models. In both sets of results the value of t_s is fixed at 700 ms and the value of $\rho_1 = 0$. In Figs. 5(a) and 6(a) coupling is turned off for the entire measurement and, as expected, there are no off-diagonal peaks. In Figs. 5(b) and 6(b) coupling is allowed during τ_1 and τ_2 but is turned off during t_s . Here we see that the μ GC and two-site exchange models lead to quite different results. If the off-diagonal peaks

TABLE II. Parameter values for the two-site exchange model as applied to the μ GC model.

$V_A (\mu\text{m}^3)$	1.28×10^6
$V_B (\mu\text{m}^3)$	1.12×10^6
$R_{2A} (\text{s}^{-1})$	2.26
$R_{2B} (\text{s}^{-1})$	7.46×10^{-2}
$R_{1A} = R_{1B}$	0
$k (\mu\text{m}^3 \text{s}^{-1})$	5.40×10^5

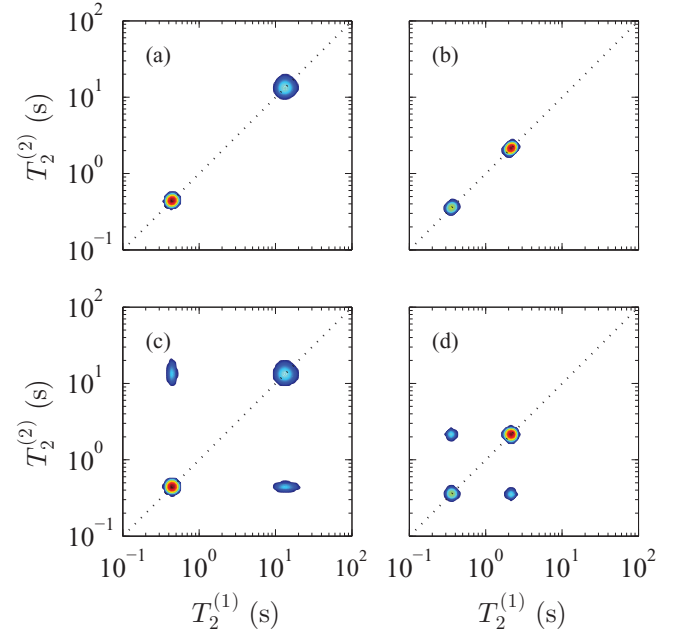


FIG. 5. (Color online) Four T_2 - T_2 calculations based on the two-site exchange model with t_s is held fixed at 700 ms. The values of k are (a) $k = 0$ at all times, (b) $k = 0$ during t_s , (c) $k = 0$ during τ_1 and τ_2 , and (d) $k = 5.40 \times 10^5 \mu\text{m}^3/\text{s}$ at all times. [Panel (d) is identical to Fig. 3(c).] All other parameters are listed in Table II. Note the qualitative agreement between these results and those in Fig. 6 except for (b).

really describe diffusion between the pores during t_s then we need to understand why they are still present in the μ GC calculations. In Figs. 5(c) and 6(c) coupling is allowed only during t_s and in Figs. 5(d) and 6(d) coupling is allowed over the entire measurement (as in Figs. 3 and 4). For both Figs. 5(c) and 6(c) and Figs. 5(d) and 6(d) the two models are in good qualitative agreement.

We note that both models share one qualitative feature: When the coupling is turned on there appears to be a much more significant reduction in the longer-lived lifetime than in the shorter one. For the two-site exchange model there is a very simple explanation, which we defer to Sec. IV A. For the μ GC model the spins in the macropore see an enhanced avenue for decay into the micropores when the barrier is removed, thereby reducing the lifetime of this mode. Most of the spins in the micropores are buried deep inside the micrograin and never see the macropore before they decay. Hence this mode’s relaxation time is largely unaffected.

In Sec. IV the behavior of the two-site exchange and μ GC models will be analyzed in detail. It is not difficult, however, to understand qualitatively what is going on. The two-site exchange model builds in the assumption that both pore populations are in the fast diffusion limit discussed in Sec. II C. Thus the magnetization is uniform within each pore between $t = 0$ and $t = \tau_1$ and then remains uniform within each pore during $\tau_1 \leq t \leq \tau_1 + t_s$. By contrast, in the μ GC model the value $\rho_2 = 2 \mu\text{m/s}$ corresponds to an intermediate case, viz., the small pores are in the fast diffusion limit but the large pores are not fully in that limit ($\rho_2 a/D_0 \approx 0.1$). Thus, as indicated in Fig. 7, between $t = 0$ and $t = \tau_1$ the

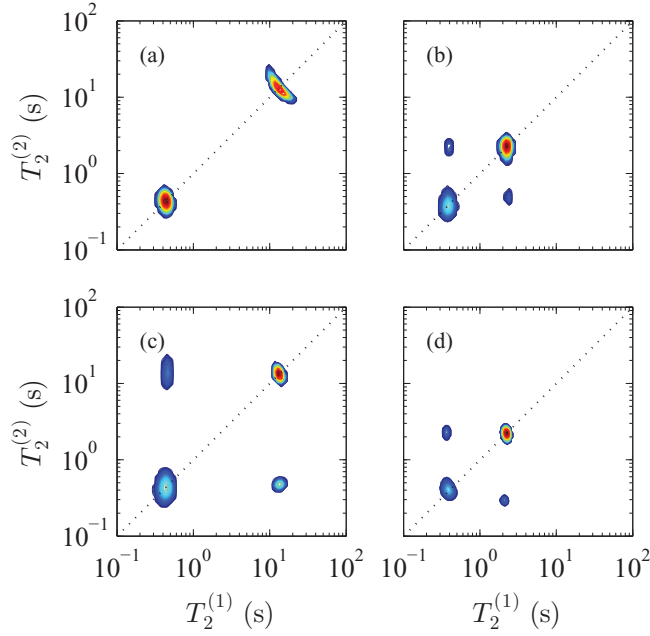


FIG. 6. (Color online) Four T_2 - T_2 simulations based on the μ GC model with t_s held fixed at 700 ms: (a) a barrier blocks coupling between the macropores and micropores for all times, (b) the barrier is in place only during t_s , (c) coupling of the two populations is allowed only during t_s , and (d) coupling is allowed at all times. [Panel (d) is identical to Fig. 4(c).] As before, the value of $\rho_2 = 2.0 \mu\text{m/s}$.

magnetization in the large pores builds up significant spatial gradients. The spin density will be largest near the center of the pore and smallest near the interface with the micropores where relaxation is taking place. During the interval $\tau_1 \leq t \leq \tau_1 + t_s$ diffusion *within* the large pores will make the magnetization somewhat more uniform (i.e., smaller in the center of the pore and larger at the interface with the micropores). Thus $M(\mathbf{r}, \tau_1)$ and $M(\mathbf{r}, \tau_1 + t_s)$ will have distinctly different representations in terms of the transverse eigenfunctions. This difference is a consequence of intrapore diffusion within the macropores rather than interpore diffusion and it is responsible for the off-diagonal peaks in the T_2 - T_2 cross plot.

To test this explanation, we have done two calculations based on the μ GC model. First, one might suppose that with $\rho_1 = 0$ and a barrier in place, the magnetization decay for $t \geq \tau_1 + t_s$ would simply be identical to that obtained from a T_2 calculation without any waiting period. From the results

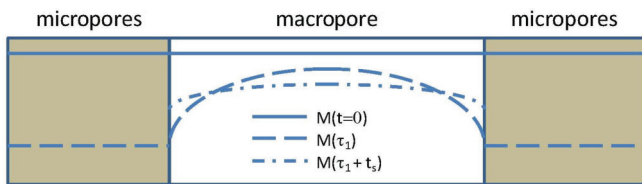


FIG. 7. (Color online) One-dimensional representation of a macropore coupled on the left and right to microporous regions. The magnetization profiles at $t = 0$, $t = \tau_1$, and $t = \tau_1 + t_s$ are indicated schematically. Because diffusion between the macropores and micropores is forbidden between $t = \tau_1$ and $\tau_1 + t_s$, the macropore magnetization at the interface *increases* during this period.

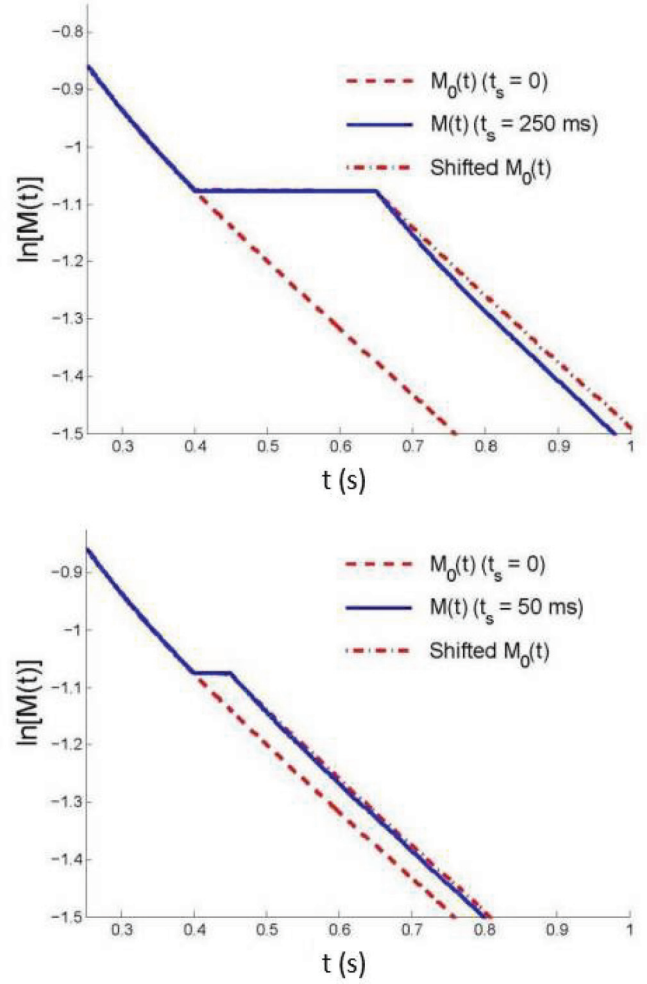


FIG. 8. (Color online) Two simulations (with different values of t_s) based on the μ GC model are compared. The dashed curves are simply the magnetization decay with coupling allowed at all times and no storage period. The solid curves are the magnetization decay with a barrier in place during t_s and storage times of 250 and 50 ms. Also shown (as dot-dashed curves) are the dashed curves shifted by t_s for $t \geq \tau_1 + t_s$. With $t_s = 250$ ms, we see a faster decay rate beginning at $t = \tau_1 + t_s$. This is a consequence of the increased magnetization at the interface pictured in Fig. 7. When $t_s = 50$ ms there is not enough time for the magnetization at the interface to adjust and there is no increase in the decay rate.

shown in Fig. 8, we see that this is not the case (although it is nearly true as $t_s \rightarrow 0$). Because the magnetization density at the interface has increased during the waiting period, the decay rate at $t = \tau_1 + t_s$ is larger than the prediction of the $t_s = 0$ calculation. (The dependence of this effect on the value of t_s will be discussed in Sec. IV C.) Second, we have repeated our calculations (with the barrier in place during t_s) with progressively smaller values of ρ_2 . In Fig. 9 we see that as ρ_2 decreases and the magnetization within the large pores becomes more uniform, the strength of the off-diagonal peaks is greatly reduced. Of course, in the limit $\rho_2 \rightarrow 0$ the macropeaks and micropeaks on the diagonal merge and we have only a single peak whose location is determined by the model's net surface area [24]. The entire pore system has moved into the fast diffusion limit, during the T_2 process.

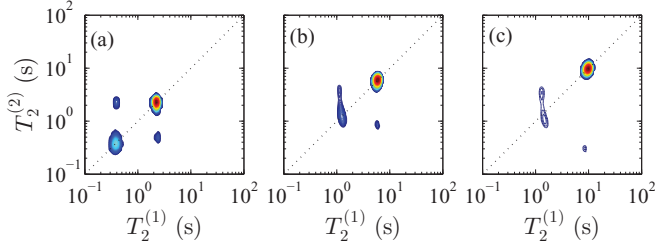


FIG. 9. (Color online) Three T_2 - t_s - T_2 simulations based on the μ GC model with progressively smaller values of ρ_2 : (a) $\rho = 2 \mu\text{m/s}$, (b) $\rho = 0.4 \mu\text{m/s}$, and (c) $\rho = 0.2 \mu\text{m/s}$. In all cases a barrier blocks coupling between the macropores and micropores during t_s . As the value of ρ_2 decreases the entire pore system moves further into the fast diffusion limit, the magnetization becomes spatially uniform at any time, and the off-diagonal peaks in the cross plot vanish.

At any instant of time during the process the magnetization becomes spatially uniform throughout the entire system.

IV. DISCUSSION

Here we examine in some detail the connection between the two-site exchange model and the results of our calculations based on the μ GC model. Is the former a useful way to understand the latter? The first question to address is whether the eigenvalues (the decay rates) can be made to match in the two systems. Second, we investigate the dependence of the matrix components of the magnetization on the storage time t_s . Finally, we look at the effects of modulating the interaction of the macropores and micropores.

A. Eigenvalues

The eigenvalues that define the behavior of the two-site exchange model are given by Eq. (17). For the T_2 part of the decay process in the μ GC model we determine the values of R_A and R_B by setting them equal to the decay rates when there is a barrier in place, i.e., when $k = 0$. For the case where $\rho_2 = 2.0 \mu\text{m/s}$ these decay rates are seen, in Fig. 6(a) or 6(c), as the reciprocals of the T_2 times where the peaks are located. From these peaks we get $R_{2A} = 1/0.4425 = 2.26 \text{ s}^{-1}$ and $R_{2B} = 1/13.4 = 0.0746 \text{ s}^{-1}$. It is straightforward to compute the volume of each of the pore subsystems in the μ GC model: $V_A = 1.28 \times 10^6 \mu\text{m}^3$ and $V_B = 1.12 \times 10^6 \mu\text{m}^3$.

There is only one remaining free parameter k in the two-site exchange model. As far as we know, there is no direct way of computing this parameter directly so we determine it by requiring that the decay rates in the coupled system match. (See the final paragraph of Sec. IV B.) In Fig. 10 we plot Eq. (17) as a function of k . The dot-dashed horizontal lines are the decay rates determined from the simulations when there is no barrier in place [Fig. 6(b) or 6(d)]. If the two-site exchange model is to have any validity for the μ GC system the former curves should intersect the latter rates at the same value of k in the two cases and we see that they basically do. We have $k = 4.96 \times 10^5 \mu\text{m}^3/\text{s}$ for the lower curve and $k = 5.80 \times 10^5 \mu\text{m}^3/\text{s}$ for the upper one. Considering the approximate way in which we have determined the decay rates from the μ GC simulations, for which there is appreciable uncertainty in the determination of the decay rates from the decay spectrum,

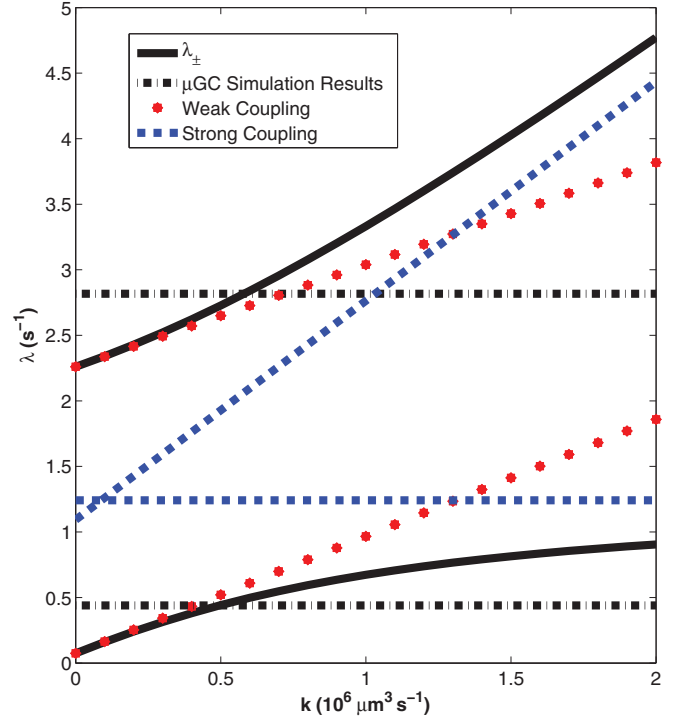


FIG. 10. (Color online) Decay rates for the two-site exchange model. The solid lines show the dependence of the lifetimes T_2 derived from Eq. (17) as a function of the coupling parameter k . The other parameters of the model are determined from the results of the μ GC simulations shown in Fig. 6(a) or 6(c), when the barrier is in place, and are given in Table II. The dot-dashed horizontal lines are the decay rates determined from the μ GC simulations in the absence of the barrier [Fig. 6(b) or 6(d)]. The intersections of these lines with the curves from Eq. (17) determine the values of k implied for each of the two modes. That these two values of k are nearly the same lends credence to the two-site exchange model as applied to the μ GC model. The dotted lines and the dashed lines are the small- k and large- k limits, respectively [Eqs. (21) and (22)].

we consider these values of k to be substantially the same. Of course, there are inherent uncertainties in this comparison. For example, the values derived from the simulations depend on (i) where exactly the barrier is placed in the simulations, (ii) the value of ρ_2 , and (iii) the details of the processing algorithm employed to determine the decay spectrum. Notwithstanding, we have found that **as long as the values of ρ_2 are not too large, the two-site exchange model is able to reproduce the two decay rates with approximately the same value of k .** We shall use the average value $k = 5.40 \times 10^5 \mu\text{m}^3/\text{s}$. The numerical values that we have determined for all the two-site exchange parameters relevant to the μ GC model are shown in Table II. These values were used to generate Figs. 3 and 5.

The dotted and dashed lines in Fig. 10 are, respectively, the small- and large- k limits described by Eqs. (21) and (22). We note in passing that the coupling in the present μ GC calculations is fairly weak in the sense that the curves shown in Fig. 10 are fairly close to Eqs. (21) at the points of intersection, i.e., $k < k_{xc}$.

We note also that we can give a simple explanation of the observation we made in Sec. III about the effect of coupling on the lifetimes that are apparent in Fig. 5. By removing

the barrier ($k = 0$ is changed to $k = 5.40 \times 10^5 \mu\text{m}^3/\text{s}$) the change in the decay rates for each of the two modes is about the same in absolute terms. However, this represents a much larger percentage change for the slower relaxing mode than it does for the faster one and this is what we see on the logarithmic axes of Fig. 5.

B. Matrix components of the magnetization

Suppose that there are only two relevant modes in the storage period and two for the T_2 part. This is rigorously true for the two-site exchange model and approximately true for the μGC model. In this section we explore the consequences of assuming it to be true for the μGC model.

At first glance Eq. (10) would lead one to think that there are six different inner products that must be evaluated in order to develop a theoretical prediction of the three functions $\mathcal{M}_{nm}$, but this is not true. We show that there are at most two independent parameters, which reduce to one for the case at hand, where we assume that there is no surface relaxivity during the storage period. Our convention in the following is that subscript 1 refers to the mode with the smaller decay rate in each case and 2 refers to the one with the larger decay rate.

Because of the completeness relation (11), one has the following result for the off-diagonal component of Eq. (10) with $n = 1$ and $m = 2$:

$$\mathcal{M}_{12}(t_s) = M_0 V_P C_{12} [e^{-\mu_1 t_s} - e^{-\mu_2 t_s}], \quad (24)$$

where

$$C_{12} = [\mathbf{m}_0 \cdot \mathbf{q}_1][\mathbf{q}_2 \cdot \mathbf{m}_0][\mathbf{q}_1 \cdot \mathbf{Q}_1][\mathbf{Q}_1 \cdot \mathbf{q}_2] / V_P \\ = -[\mathbf{m}_0 \cdot \mathbf{q}_1][\mathbf{q}_2 \cdot \mathbf{m}_0][\mathbf{q}_1 \cdot \mathbf{Q}_2][\mathbf{Q}_2 \cdot \mathbf{q}_2] / V_P. \quad (25)$$

Equation (24) exhibits the very general property that initially each off-diagonal component of $\mathcal{M}_{nm}$ increases linearly with time t_s and then reaches a maximal value before decaying to zero, as t_s is further increased. The simplicity of the two-component system lends itself to a simple geometrical interpretation. Because $[\mathbf{f} \cdot \mathbf{g}]^2 \leq [\mathbf{f} \cdot \mathbf{f}][\mathbf{g} \cdot \mathbf{g}]$ for any $\mathbf{f}, \mathbf{g}$ pair and because $|\mathbf{m}_0| = \sqrt{\mathbf{m}_0 \cdot \mathbf{m}_0} = \sqrt{V_P}$, where V_P is the total pore volume, we may define an angle ψ via

$$\mathbf{m}_0 \cdot \mathbf{q}_1 = \sqrt{V_P} \cos(\psi). \quad (26)$$

Because orthonormality and completeness of $\mathbf{q}_1$ and $\mathbf{q}_2$ imply $\mathbf{m}_0 = (\mathbf{m}_0 \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{m}_0 \cdot \mathbf{q}_2)\mathbf{q}_2$, which in turn implies $[\mathbf{m}_0 \cdot \mathbf{q}_1]^2 + [\mathbf{m}_0 \cdot \mathbf{q}_2]^2 = \mathbf{m}_0 \cdot \mathbf{m}_0 = V_P$, we have

$$\mathbf{m}_0 \cdot \mathbf{q}_2 = \sqrt{V_P} \sin(\psi). \quad (27)$$

Similarly, we may define the angle θ via

$$\mathbf{Q}_1 \cdot \mathbf{q}_1 = \cos(\theta), \quad \mathbf{Q}_1 \cdot \mathbf{q}_2 = \sin(\theta). \quad (28)$$

That is

$$\mathbf{Q}_1 = \cos(\theta)\mathbf{q}_1 + \sin(\theta)\mathbf{q}_2. \quad (29)$$

Because $\mathbf{Q}_1 \cdot \mathbf{Q}_2 = 0$ we must have

$$\mathbf{Q}_2 = \pm[-\sin(\theta)\mathbf{q}_1 + \cos(\theta)\mathbf{q}_2]. \quad (30)$$

Therefore,

$$C_{12} = \frac{1}{4} \sin(2\psi) \sin(2\theta). \quad (31)$$

Specializing to the case where the T_1 relaxivity vanishes, $\rho_1 \equiv 0$, one of the normal modes, corresponding to a uniform magnetization, does not decay at all:

$$\mu_1^0(\rho_1 = 0) \equiv 0. \quad (32)$$

The corresponding normal mode is

$$\mathbf{Q}_1(\rho_1 = 0) \equiv \mathbf{m}_0 / \sqrt{V_P}. \quad (33)$$

Now Eq. (24) reads

$$\frac{\mathcal{M}_{12}(t_s)}{M_0 V_P} = \frac{1}{4} \sin^2(2\psi) [1 - e^{-\mu_2^0 t_s}], \quad (34)$$

where we have used the fact that $\theta \equiv \psi$ when $\rho_1 = 0$ because of Eq. (33). Likewise, we may derive similar expressions for the diagonal elements. They are

$$\frac{\mathcal{M}_{11}(t_s)}{M_0 V_P} = \cos^4(\psi) + \frac{1}{4} \sin^2(2\psi) e^{-\mu_2^0 t_s}, \quad (35)$$

$$\frac{\mathcal{M}_{22}(t_s)}{M_0 V_P} = \sin^4(\psi) + \frac{1}{4} \sin^2(2\psi) e^{-\mu_2^0 t_s}. \quad (36)$$

At this level of approximation, all elements of $\mathcal{M}_{ij}$ are positive definite; as far as we know, there is no general proof that this must be so.

That all the elements of $\mathcal{M}_{nm}$ asymptotically approach constant values as $t_s \rightarrow \infty$ is a consequence of the assumption that the surface relaxivity vanishes for the storage period $\rho_1 \equiv 0$ and is independent of the assumption that there are only two relevant modes. The storage mode corresponding to uniform magnetization density $\mathbf{Q}_1 \equiv \mathbf{m}_0 / \sqrt{V_P}$ does not decay at all. Consequently, we have

$$\frac{\mathcal{M}_{nm}(t_s \rightarrow \infty)}{M_0 V_P} = A_n A_m, \quad (37)$$

where $A_{n,m}$ are given by Eq. (5). From Eq. (15) we have

$$\frac{\mathcal{M}_{nm}(t_s \rightarrow 0)}{M_0 V_P} = A_n \delta_{nm}. \quad (38)$$

Equations (34)–(38) will prove to be useful for us when we interpret our numerical results for the μGC model.

Finally, we note that the sum of all matrix elements is independent of storage time when $\rho_1 = 0$:

$$\sum_{n,m} \mathcal{M}_{nm}(t_s) = M_0 V_P. \quad (39)$$

This result is easily proved from the definition (10), using the completeness property of the T_2 modes (12). It is straightforward to demonstrate that Eqs. (34)–(36) satisfy Eq. (39).

We emphasize that Eqs. (24)–(39) all follow from two assumptions. (i) There are only two relevant modes in the problem ($\mathbf{q}_1$ and $\mathbf{q}_2$) and every quantity or function of interest ($\mathbf{m}_0, \mathbf{Q}_1, \mathbf{Q}_2$) may be expressed as a linear combination of those two. (ii) One of the modes in the storage period corresponds to uniform magnetization and does not decay at all. At this level of approximation the six different inner products implied by Eq. (10) are all expressible in terms of a single parameter ψ , which in turn is evaluated by means of its definition (26).

In order to proceed further, we need a specific model with which we evaluate ψ and μ_2^0 . Specifically, for the two-site

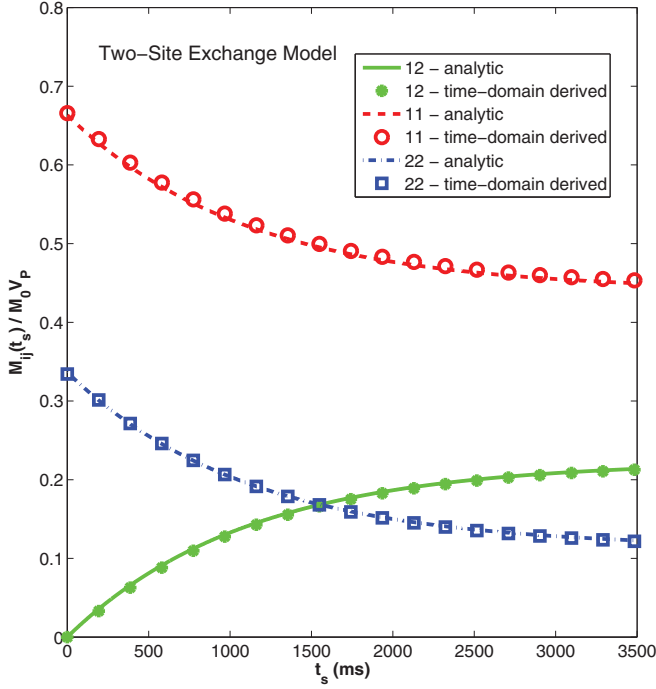


FIG. 11. (Color online) Strengths of the 2D peaks for the two-site exchange model in Fig. 3 plotted as a function of t_s . The symbols represent values numerically derived by processing time-domain data using the 2D Fredholm integral equation. The solid curves are Eqs. (34)–(36) using the parameter values listed in Table II.

exchange model we have

$$\begin{aligned}\mu_1^0 &= \lambda_-(R_A = 0, R_B = 0) = 0, \\ \mu_2^0 &= \lambda_+(R_A = 0, R_B = 0) = k(u_A + u_B),\end{aligned}\quad (40)$$

which follow from Eq. (17). Mode 1, which has a decay rate = 0, corresponds to a uniform magnetization, viz.,

$$\mathbf{Q}_1(R_A = 0, R_B = 0) \equiv \mathbf{m}_0 / \sqrt{V_P}. \quad (41)$$

The initial magnetization in turn is simply $\mathbf{m}_0 = (1, 1)$.

In Fig. 11 we plot the values of $\mathcal{M}_{ij}$ appropriate to the two-site exchange model. For each value of t_s we have generated time-domain data that we processed as if they were experimental data in the contour plots such as those shown in Fig. 3. We numerically integrate the volume under each peak as an estimate of the relevant component $\mathcal{M}_{ij}(t_s)$. These values are plotted with the relevant symbols in Fig. 11. Using the same input parameters as in Table II we plot Eqs. (34)–(36) as solid curves. Here the relevant relaxation time is given by the second of Eqs. (40): $1/\mu_2^0 = [k(u_A + u_B)]^{-1} = [\kappa_A + \kappa_B]^{-1} = 1108$ ms. We evaluate the angle ψ from its definition (26) using Eqs. (18) and (20); the initial magnetization is simply $\mathbf{m}_0 = (1, 1)$. For this parameter set we find $\psi = 0.619$ rad = 35.5° . That the strictly analytic curves agree so well with the numbers determined from the processed time-domain data provides confidence that this processing scheme will provide accurate results when we apply it to the results for the μ GC data.

To what extent do Eqs. (34)–(36) describe the evolution of the magnetization elements that we see in our μ GC simulations? We have already seen that consistency of the

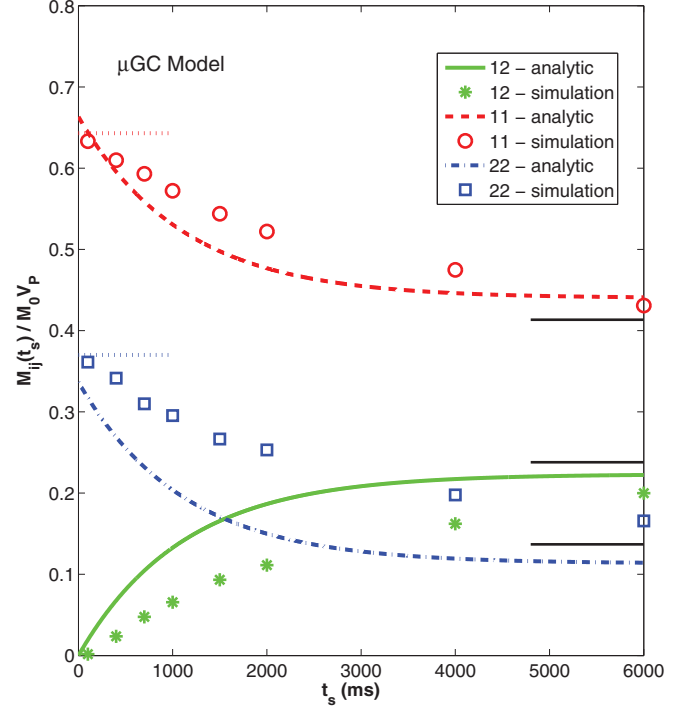


FIG. 12. (Color online) Strengths of the 2D peaks for the μ GC model in Fig. 4 plotted as a function of t_s . The symbols represent values numerically derived by processing time-domain data using the 2D Fredholm integral equation. The solid curves are Eqs. (34)–(36) using the parameter values given in Table II. The dotted lines indicate the values of the diagonal elements of the simulations at $t_s = 0$ from which the long-time asymptotic values are predicted according to Eq. (37). These values are indicated with solid horizontal lines. It is apparent that the simulation results are approaching these values in the limit of long storage times.

eigenvalues in the simulations with the predictions of the two-site exchange model determines the parameter set listed in Table II. With these values of the parameters, Eqs. (34)–(36) are replotted in Fig. 12 along with the results of the time-domain simulations based on the μ GC model, processed as before. The dotted lines indicate the values of the two diagonal elements implied by the simulations at $t_s = 0$. Using these values we indicate the asymptotic values of the simulations as predicted by Eq. (37). It is clear that the simulation data are consistent with those predictions.

We have several comments. First, the initial values predicted by the two-site exchange model for $\mathcal{M}_{11}(0)$ and $\mathcal{M}_{22}(0)$ are in rough agreement with those that are deduced from the simulation data. This is equivalent to saying that the two-site exchange model does a reasonably good job of describing ordinary T_2 decay (3) for the μ GC model. Because of Eq. (37) the behavior in the limit of long storage times is also reasonably well described by the two-site exchange model. Where the model fails is in its estimation of the time scale for equilibration, namely, the predicted decay rate $\mu_2^0 = k(u_A + u_B)$ is much larger than what we actually observe. This is a major shortcoming of the two-site exchange model in the present context. This in turn is due to the hard-wired assumption in the two-site exchange model that each subsystem is strictly in the fast diffusion limit. In reality,

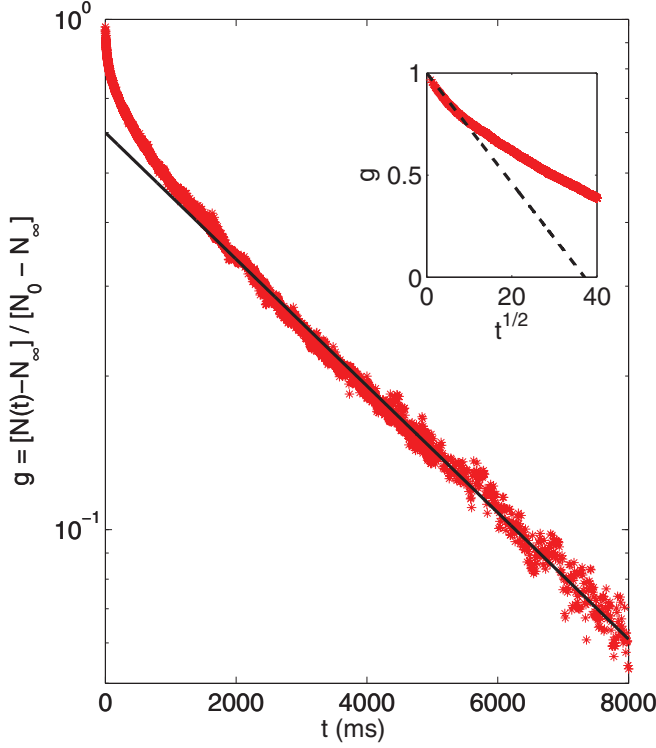


FIG. 13. (Color online) Number of walkers $N(t)$ in the macropore space monitored as a function of time for a μ GC simulation in which walkers start only in the macropore. The surface relaxivity vanishes: $\rho_1 \equiv 0$. A uniform density is achieved at long times because of diffusion. The solid line represents a decay time of 3.5 s. In the inset the data are plotted vs $\sqrt{t}$ and the dashed line corresponds to Eq. (42).

during the first T_2 decay period gradients in the magnetization are established across the entire unit cell and they relax much more slowly during the storage period than the second of Eqs. (40) implies.

In order to correctly estimate this rate we have done a simple diffusion simulation in which $\rho = 0$. We randomly launch walkers throughout the macropore only and we monitor the number of walkers within the macropore as a function of time (Fig. 13). This quantity decays from its initial value N_0 to an equilibrium value $N_\infty = N_0 V_B / (V_A + V_B)$, where V_A and V_B are the micropore and macropore volumes, respectively. Although we see several distinct decay regimes, the slowest decay rate corresponds to a lifetime of 3500 ms. We suppose that this corresponds to the dominant mode that we observe during the storage part of the T_2 - t_s - T_2 simulations in Fig. 12. Accordingly we have replotted Eqs. (34)–(36) against the numerical data for the μ GC model in Fig. 14 using this relaxation time, i.e., $\mu_2^0 = 1/3.5 \text{ s}^{-1}$. The parameter ψ in Eqs. (34)–(36) is of course unchanged because its value is set by an analysis of the coupling of the initial uniform magnetization to the T_2 modes, via Eq. (26). Although agreement is not perfect, it is much improved over that in Fig. 12.

We conclude that for this sort of μ GC geometry with the sizes approximately equal to $100 \mu\text{m}$ and relaxivity values approximately equal to $1 \mu\text{m/s}$, which are equal to those we have used in our calculations, the two-site exchange model

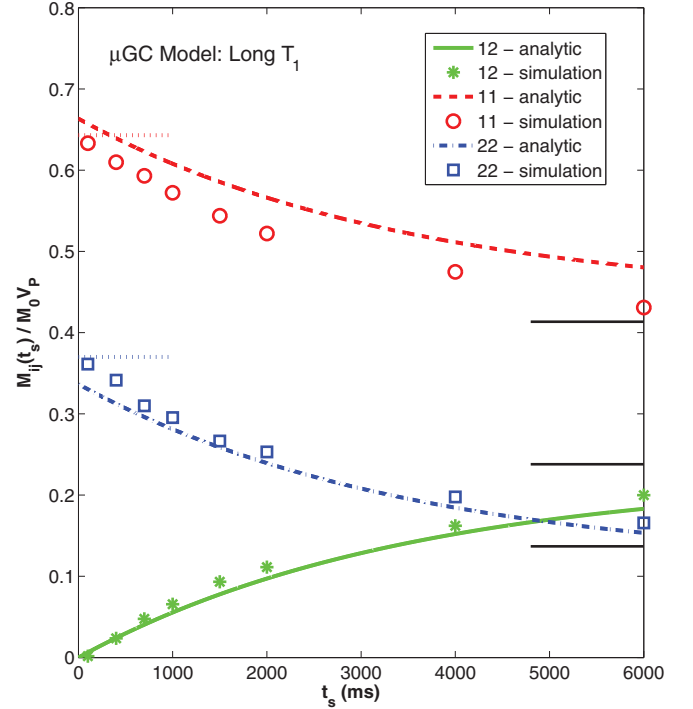


FIG. 14. (Color online) Same as for Fig. 12 except that the decay time has been changed to $T_1 = 3500 \text{ ms}$, as described in the text.

works reasonably well in describing the T_2 decay. Within the context of the two-site exchange model, the system is in the weak coupling limit $k < k_{xc}$. It fails during the storage part because we have set $\rho_1 \equiv 0$. Now the two-site exchange model seems to be in the strong coupling limit because $k \gg k_{xc} \equiv 0$. The only mechanism for decay of magnetization imbalance is via diffusion, which the two-site exchange model attributes solely to diffusion between macropores and micropores. The results of our simulations show that intrapore diffusion is clearly the limiting process in such a case. The strong suggestion of our results thus far is that if a more realistic value of $\rho_1 \approx \rho_2/2$ were used, the results of the numerical simulations would be reasonably well described by the two-site exchange model. An analysis of this case is beyond the scope of the present paper.

In connection with Fig. 13, there is an additional observation we should make. We said above that there is no direct relation between the μ GC simulations and the coupling parameter k of the two-site exchange model. One might have thought that the initial rate at which walkers diffuse into the micropores in this simulation could be interpreted as an effective exchange rate. According to Eq. (16), $\frac{dS_B}{dt}|_{t=0} = -(k/V_B)S_B|_{t=0}$, where we identify $S_B(t)$ with $N(t)$. In fact, we see that the initial slope in Fig. 13 is infinite. At short times the decay varies as $t^{1/2}$ and is well described by the equation

$$\frac{N(t)}{N_0} = 1 - \frac{S_B}{V_B} \frac{2\phi_\mu}{\alpha_\mu^{1/2} + \phi_\mu} \sqrt{D_0 t}, \quad (42)$$

where N_0 is the initial number of walkers in the macropore volume V_B , S_B is the surface area bounding the macropore, ϕ_μ is the intrinsic porosity of the micropores, and $\alpha_\mu \approx \phi_\mu^{-1/2}$

is the corresponding tortuosity. Equation (42) follows from the approximation that the micropores are describable as a continuum having a diffusivity D_0/α_μ and a porosity ϕ_μ . At short times the curvature of the interface between the macropore and this continuum may be neglected; the solution to the one-dimensional heat equation given by Carslaw and Jaeger [36] may be integrated over the coordinate perpendicular to the interface to give Eq. (42). Equation (42) is reminiscent of a result derived by Mitra *et al.* [37] in a similar but different context.

C. Effect of barriers to coupling

A more striking demonstration of the shortcomings of the two-site exchange model was presented in Figs. 6(b) and 8. There $M(\mathbf{r}, \tau_1)$ and $M(\mathbf{r}, \tau_1 + t_s)$ are not identical and so the off-diagonal peaks are not zero as would be implied by the two-site exchange model. Within the context of Eq. (10) we may write the expression for the off-diagonal element when there is a barrier in place during storage as

$$\begin{aligned} \mathcal{M}_{12}(t_s) = & M_0[\mathbf{m}_0 \cdot \mathbf{q}_1][\mathbf{q}_1 \cdot \mathbf{Q}_1][\mathbf{Q}_1 \cdot \mathbf{q}_2][\mathbf{q}_2 \cdot \mathbf{m}_0] \\ & + M_0[\mathbf{m}_0 \cdot \mathbf{q}_1][\mathbf{q}_1 \cdot \mathbf{Q}_2][\mathbf{Q}_2 \cdot \mathbf{q}_2][\mathbf{q}_2 \cdot \mathbf{m}_0] \\ & + M_0[\mathbf{m}_0 \cdot \mathbf{q}_1][\mathbf{q}_1 \cdot \mathbf{Q}_3][\mathbf{Q}_3 \cdot \mathbf{q}_2][\mathbf{q}_2 \cdot \mathbf{m}_0]e^{-\mu_3 t_s} \\ & + \dots \end{aligned} \quad (43)$$

Here $\mathbf{Q}_1$ represents the storage mode corresponding to uniform magnetization in the macropore. Because $\rho_1 = 0$ its decay rate is zero. Similarly, $\mathbf{Q}_2$ represents uniform magnetization in the micropores, whose decay rate is also zero. All the normal modes relevant to the storage period are determined by solutions to the diffusion equation with Neumann boundary conditions on the pore walls. Further, $\mathbf{Q}_3$ is the mode with the smallest decay rate μ_3 contributing to the relaxation pictured schematically in Fig. 7. There is no counterpart to it in the two-site exchange model. The lowest nontrivial decay rate may be estimated as $\mu_3 = D_0(2\pi/L)^2$, where L is the diagonal distance from one grain surface across the macropore to the other: $L = \sqrt{3}(200) - 2(125) = 96 \mu\text{m}$. This implies a decay time $1/\mu_3 = 120 \text{ ms}$. Thus, for storage times t_s significantly less than this value the magnetization does not have enough time to redistribute and contribute to $\mathcal{M}_{12}$, whereas for longer times it does. This is just the behavior we observe in our results depicted in Figs. 6(b) and 8.

V. CONCLUSION

The T_2 - t_s - T_2 measurements are an important class of 2D NMR experiments and have been used to study a wide range of complex materials. The general features of T_2 - T_2 cross plots can be modeled either by analytic calculations or by random walk simulations. In the limit of weak surface relaxation (often referred to as the fast diffusion limit) off-diagonal peaks in the cross plot are the consequence of diffusive exchange during the storage time t_s . This is the case for materials like cements, where all the pores are in the fast diffusion limit and are well mixed when exchange occurs [9,10]. More generally, for intermediate values of surface relaxation, the off-diagonal peaks are related to both intrapore and interpore diffusion and the interpretation of the measurement is more complicated. This is most likely the case in carbonate rocks (and some sandstones), where macropores are 10–100 μm in diameter, while the micropores are in the 0.1–1.0 μm range.

In order to clarify the physics underlying the T_2 - t_s - T_2 measurement in mixed pore-size systems, we presented here the results of numerical simulations for the μGC model. This model consists of a periodic arrangement of macrograins (and pores) in which each macrograin is itself porous on a length scale 10–100 times smaller than that of the macropores. It was because we used a variable step size algorithm that we were able to simulate accurately processes that occur on vastly different time scales. In order to simplify our analysis, we limited our attention to the case where, during the storage period, the surface relaxivity ρ_1 vanishes. The resulting T_2 - T_2 contour plot, at each value of the storage time t_s , was well described in terms of a simplified 2×2 matrix $\mathcal{M}_{ij}(t_s)$. We compared the results of $\mathcal{M}_{ij}(t_s)$ deduced from our simulations against those predicted directly from an analytic two-site exchange model, for which we independently deduced the values of all the relevant parameters. Although the general features of $\mathcal{M}_{ij}(t_s)$ are captured by the two-site model it does not accurately describe the time over which the matrix elements decay from their initial values to their asymptotic values. Within the context of the two-site model this decay time is simply related to the exchange parameter k . Conversely, if one were to attempt to estimate k from the observed relaxation time constant, one would deduce quite the wrong answer. The issue is that the two-site model attributes relaxation solely to interpore coupling, whereas the relaxation seen during the storage period in our simulations is due to both interpore and intrapore diffusion.

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- [1] M. D. Hürlimann, L. Venkataramanan, and C. Flaum, *J. Chem. Phys.* **117**, 10223 (2002).
 - [2] M. D. Hürlimann and L. Venkataramanan, *J. Magn. Reson.* **157**, 31 (2002).
 - [3] P. T. Callaghan, S. Godefroy, and B. N. Ryland, *J. Magn. Reson.* **162**, 320 (2003).
 - [4] L. Zielinski, R. Ramamoorthy, C. C. Minh, K. A. Al Daghar, R. H. Sayed, and A. F. Abdelaal, in *Proceedings of the SPE Annual Technical Conference and Exhibition, Florence, 2010* (SPE, Richardson, TX, 2010).
 - [5] Y. Q. Song, L. Venkataramanan, M. D. Hürlimann, M. Flaum, P. Frulla, and C. Straley, *J. Magn. Reson.* **154**, 261 (2002).
 - [6] K. E. Washburn and P. T. Callaghan, *Phys. Rev. Lett.* **97**, 175502 (2006).
 - [7] M. Fleury and J. Soualem, *J. Colloid Interface Sci.* **336**, 250 (2009).
 - [8] J. Mitchell, J. D. Griffith, J. H. P. Collins, A. J. Sederman, L. F. Gladden, and M. L. Johns, *J. Chem. Phys.* **127**, 234701 (2007).
 - [9] P. J. McDonald, J. P. Korb, J. Mitchell, and L. Monteilhet, *Phys. Rev. E* **72**, 011409 (2005).

- [10] L. Monteilhet, J. P. Korb, J. Mitchell, and P. J. McDonald, *Phys. Rev. E* **74**, 061404 (2006).
- [11] K. M. Song, J. Mitchell, H. Jaffel, and L. F. Gladden, *J. Phys. D* **45**, 105302 (2012).
- [12] D. W. Weber, J. Mitchell, J. McGregor, and L. F. Gladden, *J. Phys. Chem. C* **113**, 6610 (2009).
- [13] J. D. Griffith, J. Mitchell, A. E. Bayly, and M. L. Johns, *J. Phys. Chem. B* **113**, 7156 (2009).
- [14] P. T. Callaghan, S. Godefroy, and B. N. Ryland, *Magn. Reson. Imag.* **21**, 243 (2003).
- [15] P. Callaghan and I. Furó, *J. Chem. Phys.* **120**, 4032 (2004).
- [16] Y. Q. Song, *Prog. Nucl. Magn. Reson. Spectrosc.* **55**, 324 (2009).
- [17] J. Cox, P. J. McDonald, and B. A. Gardiner, *Holzforschung* **64**, 259 (2010).
- [18] J. Lee, C. Labadie, C. Springer, and G. Harbison, *J. Am. Chem. Soc.* **115**, 7761 (1993).
- [19] T. C. Chandrasekera, J. Mitchell, E. J. Fordham, L. F. Gladden, and M. L. Johns, *J. Magn. Reson.* **194**, 156 (2008).
- [20] J. R. Zimmerman and W. E. Brittin, *J. Phys. Chem.* **61**, 1328 (1957).
- [21] H. M. McConnell, *J. Chem. Phys.* **28**, 430 (1958).
- [22] K. R. Brownstein and C. E. Tarr, *Phys. Rev. A* **19**, 2446 (1979).
- [23] J. Mitchell, T. C. Chandrasekera, M. L. Johns, L. F. Gladden, and E. J. Fordham, *Phys. Rev. E* **81**, 026101 (2010).
- [24] T. S. Ramakrishnan, L. M. Schwartz, E. J. Fordham, W. E. Kenyon, and D. J. Wilkinson, *Log Anal.* **40**, 260 (1999).
- [25] M. Knackstedt, A. Carnerup, A. Golab, R. Sok, B. Young, and L. Riepe, *Petrophysics* **54**, 216 (2013).
- [26] S. Ryu and D. L. Johnson, *Phys. Rev. Lett.* **103**, 118701 (2009).
- [27] H. Y. Carr and E. M. Purcell, *Phys. Rev.* **94**, 630 (1954); S. Meiboom and D. Gill, *Rev. Sci. Instrum.* **29**, 688 (1958).
- [28] M. H. Levitt, *Spin Dynamics: Basics of Nuclear Magnetic Resonance* (Wiley, Chichester, 2001).
- [29] J. Mitchell, T. C. Chandrasekera, and L. F. Gladden, *Prog. Nucl. Magn. Reson. Spectrosc.* **62**, 34 (2012).
- [30] A. N. Tikhonov and V. Y. Arsenin, *Solutions of Ill-Posed Problems* (Winston, Washington, DC, 1977).
- [31] Y. Q. Song, L. Zielinski, and S. Ryu, *Phys. Rev. Lett.* **100**, 248002 (2008).
- [32] S. Rodts and D. Bytchenkoff, *J. Magn. Reson.* **205**, 315 (2010).
- [33] D. Bytchenkoff and S. Rodts, *J. Magn. Reson.* **208**, 4 (2011).
- [34] G. Wahba, *SIAM J. Numer. Anal.* **14**, 651 (1977).
- [35] Z. Zhang, D. L. Johnson, and L. M. Schwartz, *Phys. Rev. E* **84**, 031129 (2011).
- [36] H. S. Carslaw and J. C. Jaeger, *Conduction of Heat in Solids*, 2nd ed. (Clarendon, Oxford, 2008).
- [37] P. P. Mitra, P. N. Sen, and L. M. Schwartz, *Phys. Rev. B* **47**, 8565 (1993).